IIIT BHOPAL METAVERSE

An interactive virtual campus and real-time educational digital twin

BRANCH

CSE-AI

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A campus that can be explored, not just viewed

Browser-based 2.5D campus

Built with Phaser 3 and React.

Real-time learning

Interactive navigation, classrooms, collaborative learning, and an AI guide.

Practical digital twin

Designed for orientation and hybrid education at IIIT Bhopal.

Where conventional online learning falls short



Place is hard to find

New students and visitors need help locating campus blocks, CCF labs, departments, and services.



Presence is hard to feel

Traditional video classes provide communication but little sense of place or shared campus presence.



Rooms are poorly separated

Virtual classrooms can suffer from background-noise leakage and weak separation between rooms.



Access is hard to discover

Accessibility information about ramps and inclusive facilities is often difficult to discover.

03 · PROBLEM STATEMENT

Connect campus space, learning, and inclusion

How can a low-cost browser platform reproduce the utility of a campus while supporting safe real-time classes, intelligent navigation, collaborative teaching, and accessible discovery?

Navigable campus

Clear routes and live guidance.

Safe classrooms

Server-enforced room isolation with clean media teardown.

Inclusive learning

Role-aware tools, synchronized surfaces, and inclusive facilities.

A dual-layer virtual campus



Spatial layer

Phaser 3 2.5D campus world with tilemaps, cameras, sprites, colliders, and 60 FPS target.



Communication layer

WebRTC media streams, WebSockets, and REST APIs for real-time classroom interaction.



Deployment model

Demo mode uses bundled in-memory data; production can persist to MongoDB.

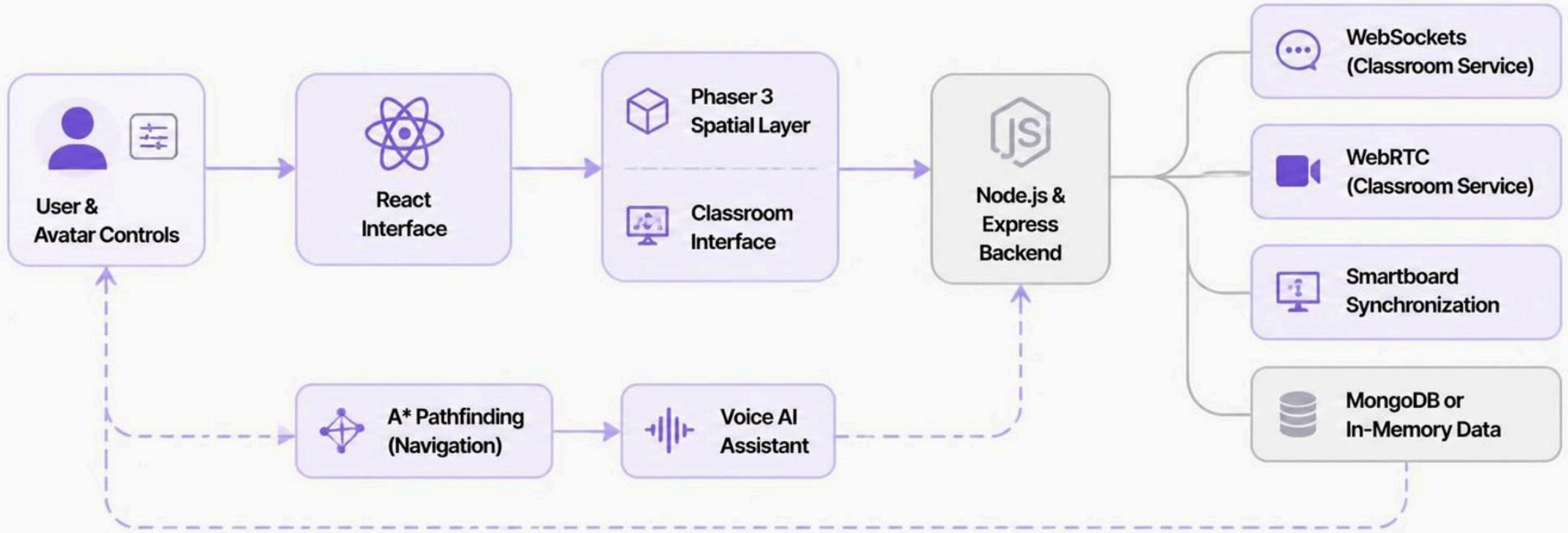


Intelligence

A* route planning, dynamic re-routing, and a voice-enabled assistant using local browser capabilities.

From user intent to campus action

A LAYERED ARCHITECTURE CONNECTS SPATIAL INTERACTION, REAL-TIME SERVICES, DATA, AND GUIDANCE.



Navigation, classrooms, and collaborative tools



Navigation

WASD/arrows, sprinting, click-to-move, distance and walking ETA, and re-routing after a 120-pixel deviation.



Classrooms

Outdoor ambience, isolated room audio, teacher/student dashboards, screen sharing, attendance export, raise-hand, chat, and Q&A.



Smartboard

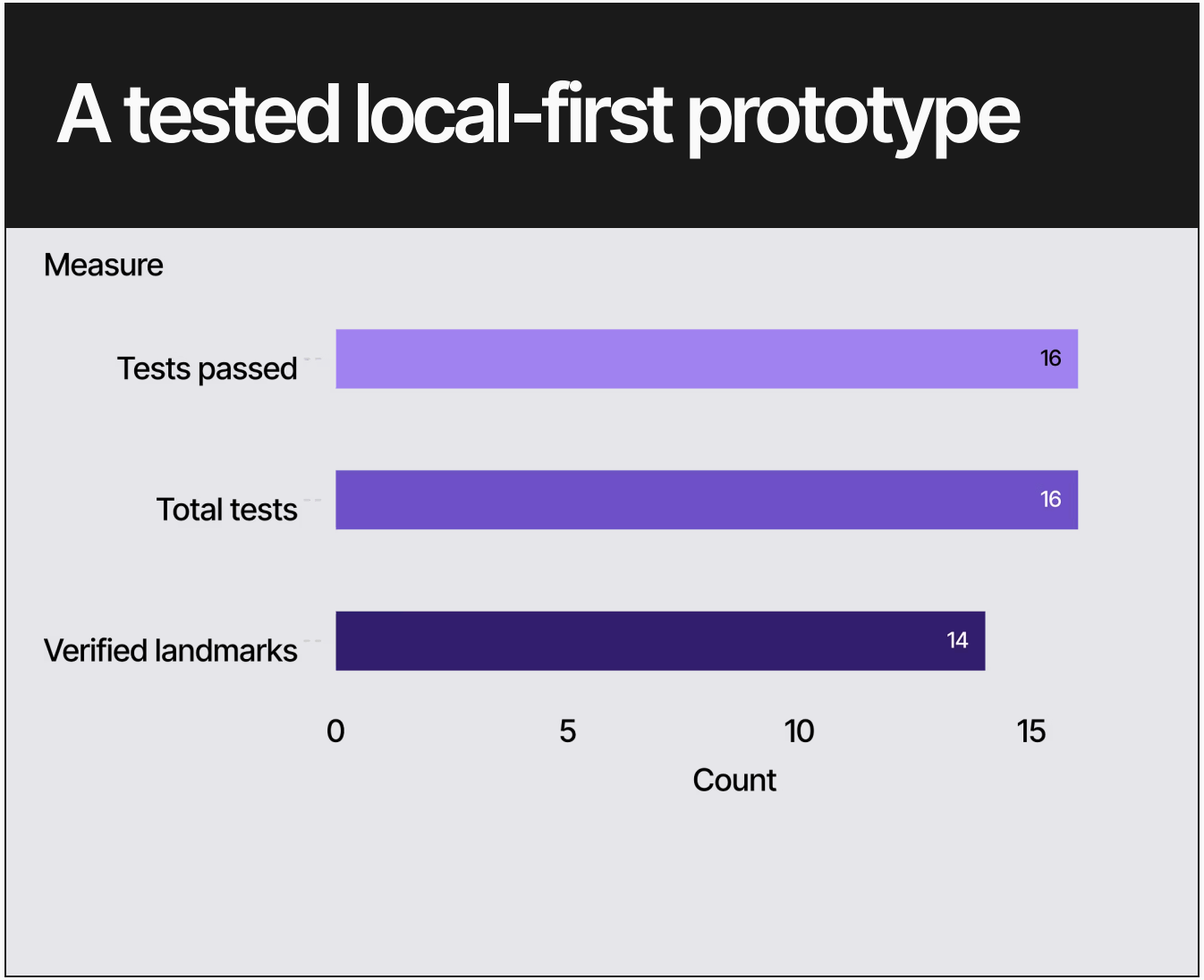
4K canvas, drawing tools, WebSocket synchronization, teacher lock, and PNG export.



AI guide

Intent classification for directions, timings, and faculty queries with Web Speech API text-to-speech.

A tested local-first prototype



ANALYSIS

Spatial orientation, presence, room-aware audio, and collaborative campus context.

Compared with conventional online classes, the system adds these capabilities while avoiding paid cloud APIs.

BROWSER-BASED CAMPUS TARGETED AT 60 FPS WITHOUT A HIGH-END GPU.

DATASET INCLUDES 14 VERIFIED INSTITUTIONAL LANDMARKS, INCLUDING THE CCF LAB, ECE BLOCK, CENTRAL LIBRARY, FOOD COURT, AND HOSTELS.

08 · CONCLUSION

A practical digital twin for campus learning

The prototype demonstrates a practical digital twin for campus orientation and hybrid classes.

Core contribution

Spatial navigation, isolated classrooms, collaborative teaching, accessibility, and local-first deployment.

Future work

Multi-avatar presence across the open campus grid and integration with official institute noticeboards.

References

No.	Source
01	IIIT Bhopal Metaverse project presentation, supplied source PDF, “IIIT Bhopal Metaverse: An Interactive Virtual Campus & Real-Time Educational Digital Twin.”
02	Project implementation technologies named in the supplied material: Phaser 3, React 18, Vite, Tailwind CSS, Node.js, Express.js, WebSockets, WebRTC, MongoDB, Web Speech API, and LiveKit Access Token Schema.